Tècniques i Eines Bioinformàtiques

Boyer-Moore Algorithm

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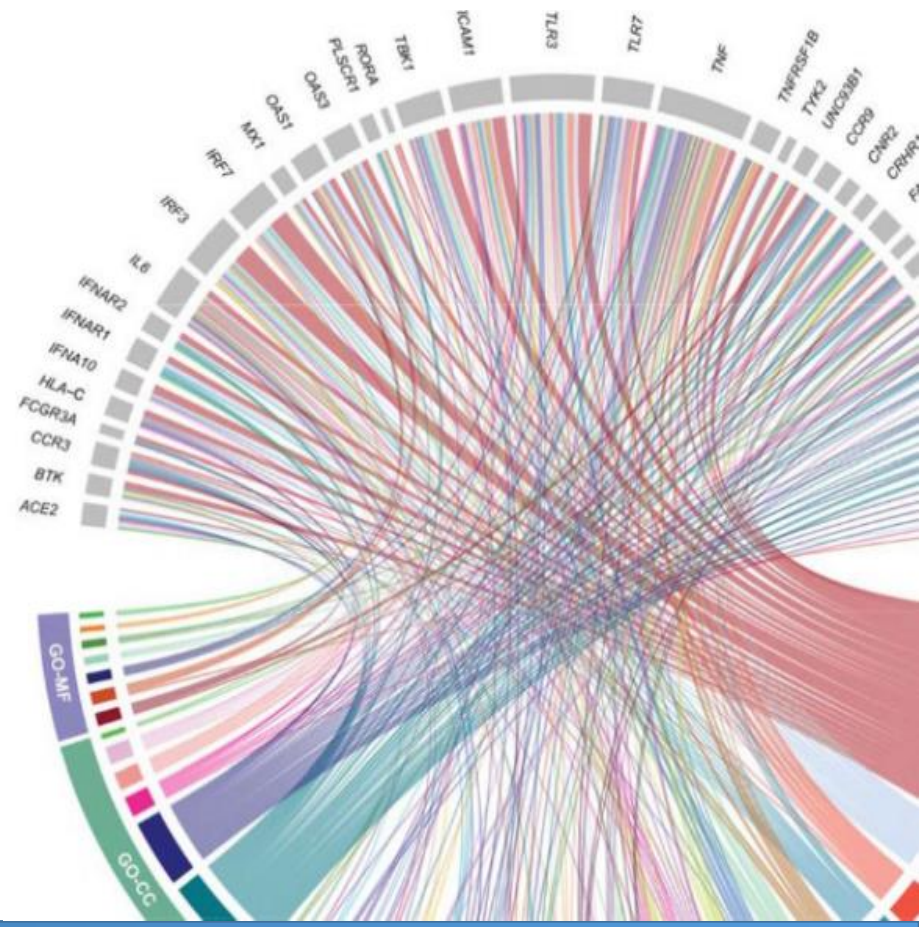


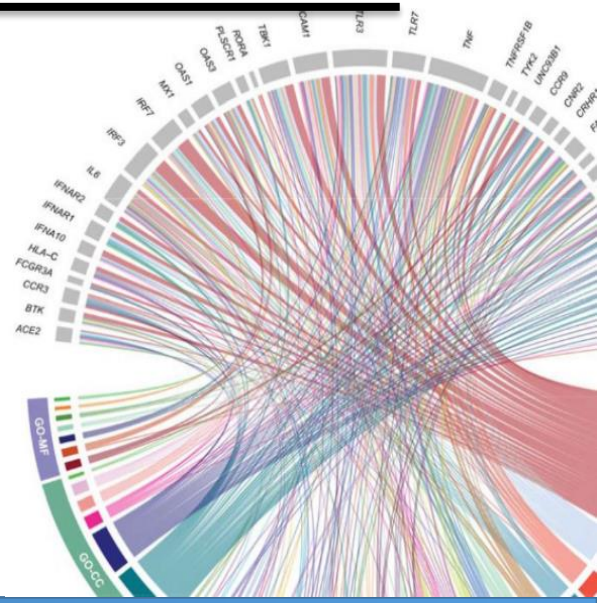
Acknowledgements

Many pictures and materials are taken from **Ben Langmead's course**.

Course heavily inspired in:

- **Genome-Scale Algorithm Design**. Veli Mäkinen, Djamal Belazzougui, Fabio Cunial, Alexandru I. Tomescu. Cambridge University Press.
- **Algorithms on Strings, Trees, and Sequences**. Dan Gusfield. Cambridge University Press.
- **An Introduction to Bioinformatics Algorithms.** Neil C. Jones, Pavel A. Pevzner. MIT Press.

1 Boyer-Moore Algorithm



Can we improve on the naïve algorithm?

```
P: word
T: There would have been a time for such a word
   ..........word ...................................................>
```

u doesn't occur in *P*, so skip next two alignments

```
P: word
T: There would have been a time for such a word
   ..........word ...................................................>
            word    skip!
             word    skip!
              word
```

Boyer-Moore

Learn from character comparisons to skip pointless alignments

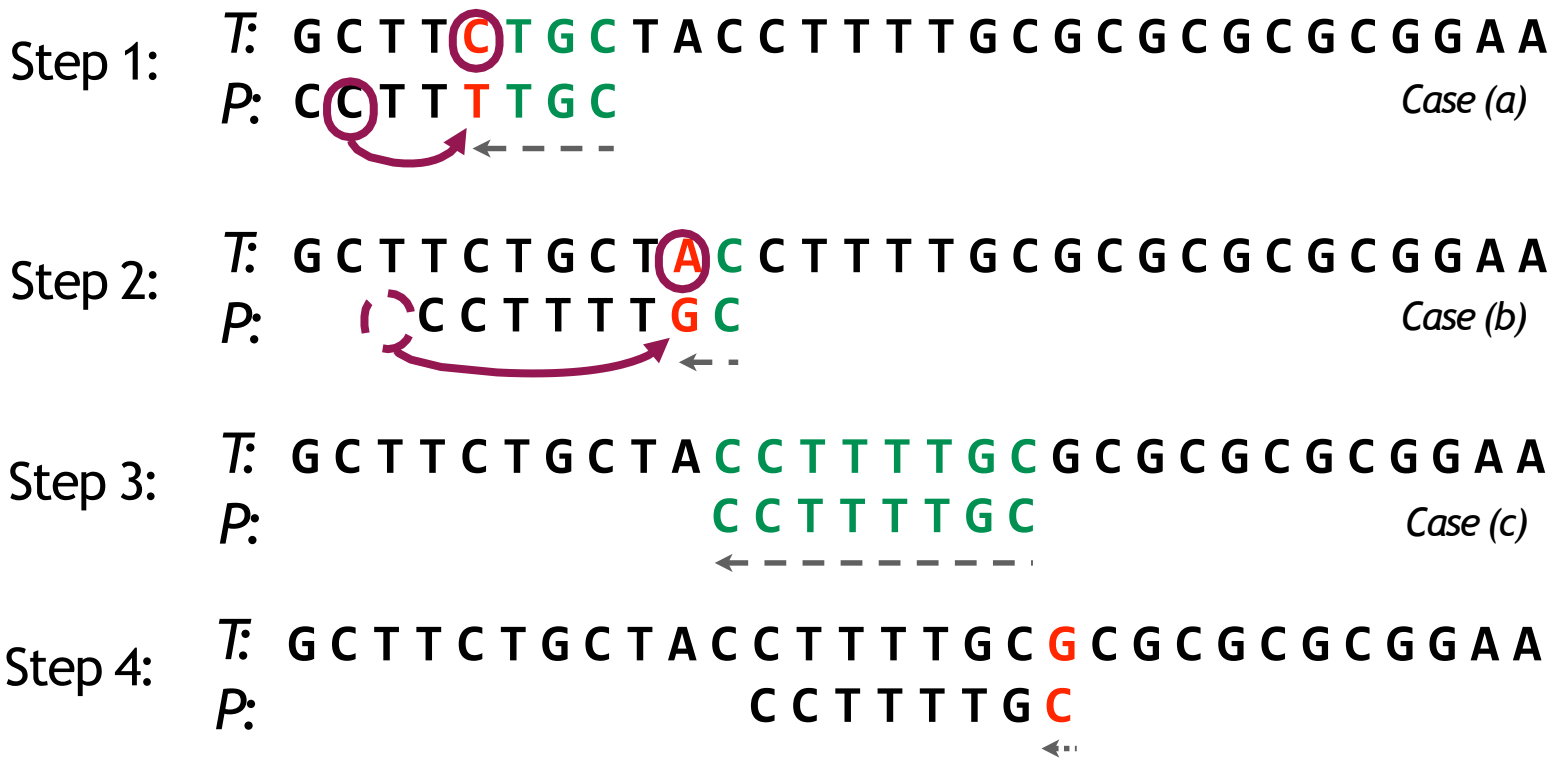
1. When we hit a mismatch, move *P* along until the mismatch becomes a match — "Bad character rule"
2. When we move *P* along, make sure characters that matched in the last alignment also match in the next alignment — "Good suffix rule"
3. Try alignments in one direction, but do character comparisons in *opposite* direction — "Longer skips"

```
P: word
T: There would have been a time for such a word
   ............word ..................................................→
              ← –
```

Boyer, RS and Moore, JS. "A fast string searching algorithm." *Communications of the ACM* 20.10 (1977): 762-772.

Boyer-Moore: Bad character rule

- Upon mismatch, skip alignments until:
  - (a) mismatch becomes a match
  - (b) P moves past mismatched character.
  - (c) If there was no mismatch, don't skip



Boyer-Moore: Bad character rule

```
Step 1:  T: G C T T C T G C T A C C T T T T G C G C G C G C G C G G A A
         P: C C T T T T G C

Step 2:  T: G C T T C T G C T A C C T T T T G C G C G C G C G C G G A A
         P:       C C T T T T G C

Step 3:  T: G C T T C T G C T A C C T T T T G C G C G C G C G C G G A A
         P:                     C C T T T T G C
```

Up to step 3, we skipped 8 alignments

5 characters in *T* were never looked at

Boyer-Moore: Good suffix rule

Let t = substring matched by inner loop; skip until (a) there are no mismatches between P and t *or* (b) P moves past t

Step 1:

```
                  t
T:  C G T G C C T A C T T A C T T A C T T A C T T A C G C G A A
P:  C T T A C T T A C
```

Step 2:

```
                      t
T:  C G T G C C T A C T T A C T T A C T T A C T T A C G C G A A
P:            C T T A C T T A C
```

Step 3:

```
T:  C G T G C C T A C T T A C T T A C T T A C T T A C G C G A A
P:                  C T T A C T T A C
```

Boyer-Moore: Good suffix rule

Let t = substring matched by inner loop; skip until (a) there are no mismatches between P and t *or* (b) P moves past t

Step 1:

```
                    t
T:  C G T G C C T A C T T A C T T A C T T A C T T A C G C G A A
P:  C T T A C T T A C
```

t occurs *in its entirety* to the left within P

Step 2:

```
                    t
T:  C G T G C C T A C T T A C T T A C T T A C T T A C G C G A A
P:            C T T A C T T A C
```

prefix of P matches a *suffix* of t

Step 3:

```
T:  C G T G C C T A C T T A C T T A C T T A C T T A C G C G A A
P:                  C T T A C T T A C
```

Case (a) has two subcases according to whether t occurs *in its entirety* to the left within P (as in step 1), or a *prefix* of P matches a *suffix* of t (as in step 2)

Boyer-Moore: Putting it together

How to *combine* bad character and good suffix rules?

T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
P: G T A G C G G C G

bad char says skip 2, good suffix says skip 7

Take the maximum! (7)

Boyer-Moore: Putting it together

Use bad character or good suffix rule, *whichever skips more*

Step 1:

T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A

P: G T A G C G G C G

bc: **6**, gs: 0 *bad character*

Step 2:

T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A

P: G T A G C G G C G

bc: 0, gs: **2** *good suffix*

Step 3:

T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A

P: G T A G C G G C G

bc: 2, gs: **7** *good suffix*

Step 4:

T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A

P: G T A G C G G C G

Boyer-Moore: Comparisons Skipped (characters ignored)

11 characters of *T* we ignored

Step 1:
T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
P: G T A G C G G C G

Step 2:
T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
P: G T A G C G G C G

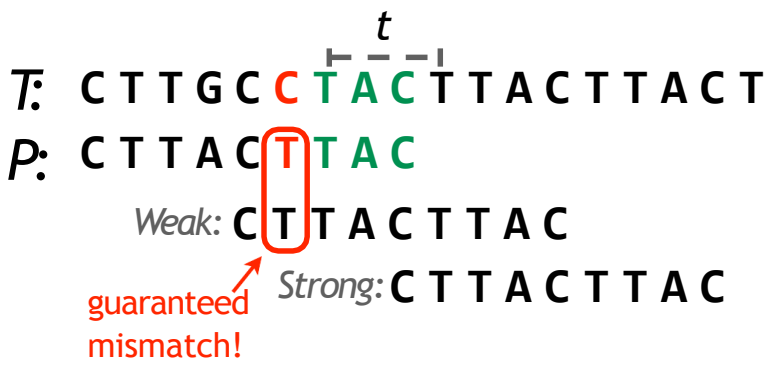
Step 3:
T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
P: G T A G C G G C G

Step 4:
T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
P: G T A G C G G C G

Skipped 15 alignments

Boyer-Moore: Good suffix rule

We learned the *weak* good suffix rule; there is also a *strong* good suffix rule



Strong good suffix rule skips more than weak, at no additional penalty

Strong rule is needed for proof of Boyer-Moore's O(n + m) worst-case time.
Gusfield discusses proof(s) in first several sections of ch. 3

Boyer-Moore implementation

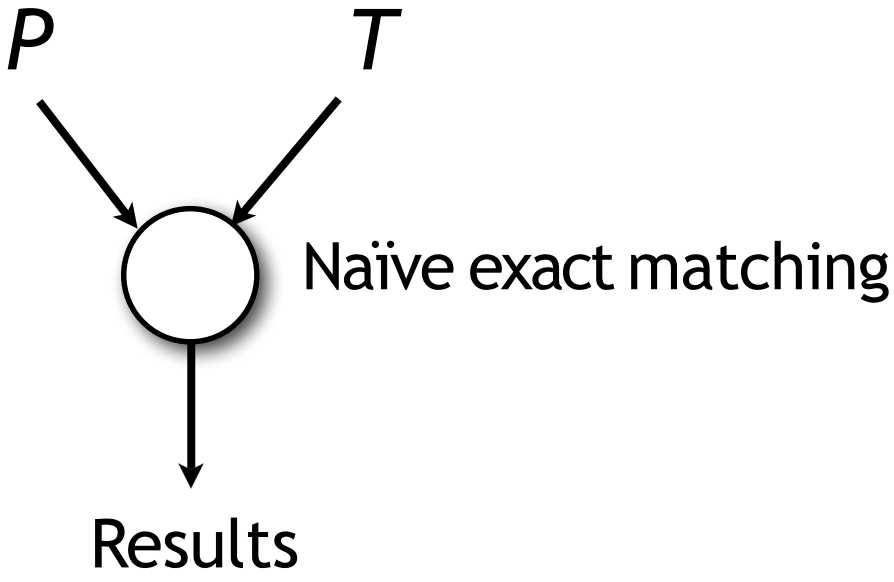
Boyer-Moore Algorithm

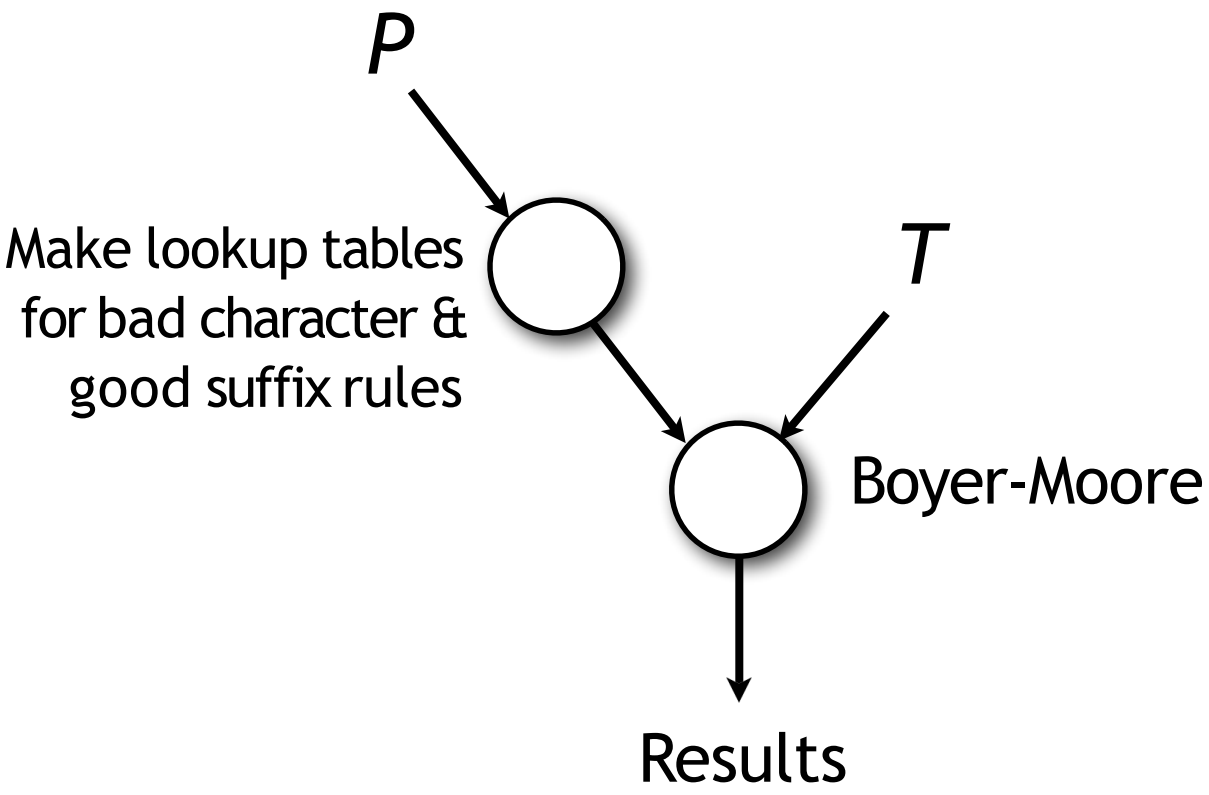
```
def boyer_moore(p,p_bm,t):
  # Do Boyer-Moore matching
  i = 0
  occurrences = []
  while i < len(t) - len(p) + 1: # Left to right
    shift = 1
    mismatched = False
    for j in range(len(p)-1,-1,-1):  #  Right  to  left
      if p[j] != t[i+j]:
        skip_bc = p_bm.bad_character_rule(j,t[i+j])
        skip_gs = p_bm.good_suffix_rule(j)
        shift = max(shift,skip_bc,skip_gs)
        mismatched = True
        break
    if not mismatched:
      occurrences.append(i)
      skip_gs = p_bm.match_skip()
      shift = max(shift,skip_gs)
    i += shift
    return  occurrences
```

Full Code:

http://j.mp/CG_BoyerMoore

Preprocessing. Naïve algorithm vs. Boyer-Moore





Boyer-Moore: Preprocessing

- Pre-calculate skips for all possible mismatch scenarios!
- This can be constructed efficiently. See Gusfield 2.2.2.
- As with bad character rule, good suffix rule skips can be precalculated efficiently. See Gusfield 2.2.4 and 2.2.5.
- For both tables, the calculations only consider P. No knowledge of T is required.

T: A A T C A A T A G C
P: T C G C

P

Σ	T	C	G	C
A	0	1	2	3
C	0	-	0	-
G	0	1	-	0
T	-	0	1	2

Boyer-Moore: Best case

What's the best case?

P: **bbbb**

T: **aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa**

bbbbbbbb bbbb bbbb bbbbbbbb

bbbbbbbb bbbb bbbb bbbb

Every alignment yields immediate mismatch and bad character rule skips *n* alignments

How many character comparisons? *floor(m / n)*

Boyer-Moore: Worst case

Boyer-Moore, with refinements in Gusfield, is $O(n + m)$ time

Given $n < m$, can simplify to $O(m)$

Is this better than naïve?

For naïve, worst-case # char comparisons is $n(m - n + 1)$

Boyer-Moore: $O(m)$, naïve: $O(nm)$

Reminder: $|P| = n$ $|T| = m$

Naïve vs Boyer-Moore

- As m and n grow, the number characters comparisons grows with...

  $|P| = n$

  $|T| = m$

	Naïve matching	Boyer-Moore
Worst case	$m \cdot n$	m
Best case	m	m / n

Performance comparison

Simple Python implementations of naïve and Boyer-Moore:

	Naïve matching		Boyer-Moore		
	# character comparisons	wall clock time	# character comparisons	wall clock time	
P: "tomorrow" **T**: Shakespeare's complete works	5,906,125	2.90 s	785,855	1.54 s	17 matches $\|T\| = 5.59$ M
P: 50 nt string from Alu repeat* **T**: Human reference (hg19) chromosome 1	307,013,905	137 s	32,495,111	55 s	336 matches $\|T\| = 249$ M

* GCGCGGTGGCTCACGCCTGTAATCCCAGCACTTTGGGAGGCCGAGGCGGG

Preprocessing: Boyer-Moore

Preprocessing: trade one-time cost for reduced work overall via *reuse*

Boyer-Moore preprocesses P into lookup tables that are ***reused***

reused for each alignment of P to T_1

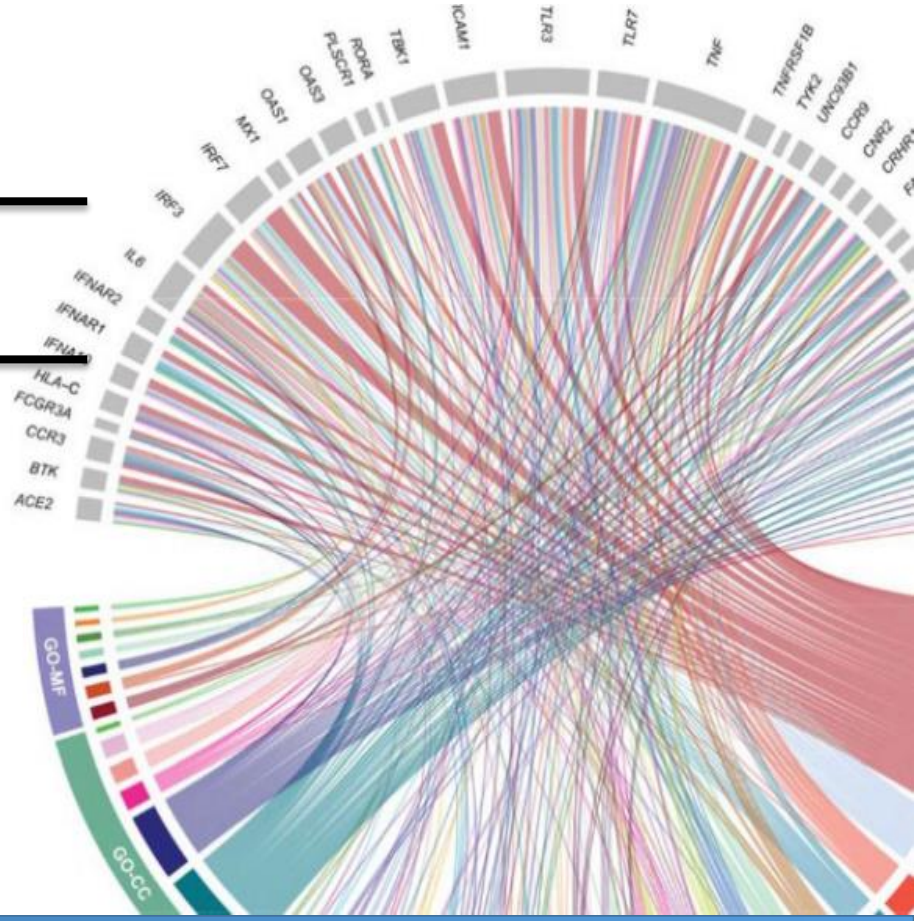
If you later give me T_2, I *reuse* the tables to match P to T_2

If you later give me T_3, I *reuse* the tables to match P to T_3

…

Cost of **preprocessing is *amortized*** over alignments & texts

Questions


ACE2
BTK
CCR3
FCGR3A
HLA-C
IFNAR1
IFNAR2
IL6
IRF3
IRF7
MX1
OAS1
OAS3
PLSCR1
RORA
TBK1
ICAM1
TLR3
TLR7
TNF
TNFRSF1B
TYK2
CCR9
CNR2
GO-MF
GO-CC